

#ASLI ENGINEERING

Talking the Redis Language

SWIPE

BY

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Talking the Redis language

For a Redis Server to understand what a client wants, the client needs to follow a protocal



Redis Serialization Protocal

RESP supports common datatypes like : int, string and arrays and a way to convey errors

Redis sends command as an array of string

eg: PUT K V is sent as ["PUT", "K", "V"]

Serialized using RESP

RESP Description

Redis uses RESP as a request - response protocol

ie. client sends request in RESP

server responds back in RESP

Every data type starts with a special character and the data ends with \r\n (CRLF)

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Simple Strings

Simple strings start with a '+'
Followed by the string 'PONG'
Followed by a CRLF '\r\n'

+PONG\r\n

Minimal overhead because it requires n+3 bytes in response

Integers

Integers start with a ':'
Followed by the integer '1729'
Followed by the CRLF '\r\n'

:1729\r\n

Bulk Strings

Bulk strings start with a '$'
Followed by the number of bytes
Followed by a CRLF '\r\n'
Followed by the actual string
Followed by the CRLF '\r\n'

$4\r\nPONG\r\n